

KINGSTON CLIMATE, ON, Canada

WMO: 718200

Lat: 44.2232N

Lon: 76.5994W

Elev: 305

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-1.3	-16.1	2.3	-5.8	-10.6	3.1	0.5	30.8	28.3	27.7	28.7	7.8	20	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	82.3	71.0	79.6	70.1	77.7	69.3	74.4	79.0	72.9	77.0	71.5	75.8	11.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	123.5	77.2	71.5	117.6	75.9	69.9	111.3	74.8	38.2	79.2	36.8	77.1	35.5	75.9	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	21.8	19.4	-16.3	88.0	5.5	3.4	-20.2	90.4	-23.5	92.4	-26.6	94.3	-30.6	96.7
			-16.7	77.3	5.5	1.9	-20.6	78.7	-23.8	79.7	-26.9	80.8	-30.8	82.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	20.0	22.0	30.6	42.4	55.6	63.2	69.8	68.5	61.4	49.7	38.0	27.9
	DBStd	19.20	11.92	10.80	9.81	7.13	7.20	5.22	4.88	4.78	7.00	7.64	8.66	10.53
	HDD50	3750	931	785	603	244	30	0	0	0	3	100	368	686
	HDD65	7359	1396	1205	1066	679	299	95	12	21	150	474	811	1151
	CDD50	2247	0	0	3	15	205	395	613	573	346	91	6	0
	CDD65	383	0	0	0	0	9	40	160	130	43	1	0	0
	CDH74	1672	0	0	0	0	40	150	801	525	155	1	0	0
	CDH80	207	0	0	0	0	4	12	132	48	11	0	0	0
Wind	WSAvg	9.4	10.4	10.0	9.9	10.1	8.3	7.9	8.1	8.4	8.7	10.2	10.5	10.6
Precipitation	PrecAvg	39.5	2.9	2.6	3.1	3.2	3.2	3.0	2.5	3.4	3.8	3.5	4.0	4.1
	PrecMax	48.1	7.5	5.4	5.6	6.2	5.4	5.4	6.1	7.8	7.3	5.7	6.1	8.3
	PrecMin	31.3	1.0	0.6	1.3	1.3	1.3	1.1	0.7	1.3	1.1	1.0	2.0	1.9
	PrecStd	4.2	1.6	1.5	1.2	1.5	1.2	1.3	1.4	1.5	1.7	1.3	1.2	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	46.2	59.2	66.5	78.0	81.8	88.3	84.4	81.4	70.5	59.3	52.3
		MCWB	43.5	41.9	54.5	55.9	64.1	67.9	73.7	71.2	72.2	63.8	54.3	49.9
	2%	DB	41.4	42.1	50.5	60.9	74.2	78.1	83.4	80.8	78.0	67.3	55.9	47.4
		MCWB	39.8	39.2	44.2	50.3	62.5	66.8	71.8	70.7	70.8	62.2	51.5	45.2
	5%	DB	38.4	38.8	46.2	56.6	70.7	75.1	80.5	78.6	75.5	64.5	53.6	44.6
		MCWB	36.3	36.0	41.7	47.6	61.1	65.5	70.6	70.3	69.7	59.6	49.9	42.7
	10%	DB	35.9	36.5	42.7	53.2	67.2	72.5	78.2	76.9	73.1	61.9	51.1	41.7
		MCWB	33.8	34.0	39.0	46.3	58.9	63.9	69.6	69.5	67.7	57.6	47.5	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.7	43.3	54.1	58.3	68.1	72.8	77.4	75.6	74.9	66.6	55.5	49.9
		MCDB	44.4	44.8	57.5	64.0	74.3	78.4	82.5	80.9	79.5	68.7	59.1	51.8
	2%	WB	40.2	39.7	46.0	53.1	65.2	69.6	74.9	73.9	72.5	63.0	52.3	45.6
		MCDB	41.2	42.0	48.8	57.5	71.2	74.8	79.9	77.8	76.1	65.8	55.3	47.0
	5%	WB	36.6	36.5	42.0	50.0	62.9	67.4	73.3	72.6	70.8	60.7	50.3	42.7
		MCDB	38.1	38.5	45.4	54.3	68.3	72.5	77.7	76.4	74.3	63.8	53.0	44.5
	10%	WB	34.1	34.1	39.2	47.2	60.3	65.7	71.7	71.2	68.7	57.8	48.1	39.4
		MCDB	35.8	36.0	42.7	51.8	66.0	70.2	76.4	75.2	72.5	61.4	51.0	41.5
Mean Daily Temperature Range	5% DB	MDBR	14.8	14.3	13.9	15.4	17.0	15.3	15.4	15.0	16.3	14.0	13.3	12.0
		MCDBR	13.4	14.0	17.0	21.5	21.0	19.0	17.3	15.8	16.2	14.9	15.8	13.4
		MCWBR	13.0	12.7	12.9	14.1	12.6	11.1	10.1	9.6	11.0	11.4	13.4	13.3
	5% WB	MCDBR	12.9	14.0	15.3	18.3	17.9	15.1	14.7	12.7	14.7	13.6	14.0	13.4
		MCWBR	12.9	13.2	12.6	14.1	12.4	10.1	9.5	9.0	11.2	11.6	12.7	13.5
Clear-Sky Solar Irradiance	taub	0.292	0.297	0.322	0.355	0.383	0.421	0.426	0.417	0.372	0.350	0.308	0.284	
	taud	2.407	2.394	2.431	2.397	2.372	2.288	2.282	2.304	2.412	2.463	2.525	2.493	
	Ebn at Noon	264	285	291	288	282	270	267	266	270	261	258	258	
	Edh at Noon	23	29	31	36	38	41	41	39	32	27	21	20	
All-Sky Solar Radiation	RadAvg	409	682	1107	1422	1725	1833	1868	1643	1311	780	482	326	
	RadStd	42	64	95	158	167	146	155	83	108	70	58	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page